



Alberta Public Health Disease Management Guidelines

Multisystem Inflammatory Syndrome in Children
and Adolescents (MIS-C)

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Health and Wellness Promotion Branch

Public Health and Compliance Branch

Alberta Health

Multisystem Inflammatory Syndrome in Children and Adolescents (MIS-C) Public Health Disease Management Guideline | Alberta Health, Government of Alberta

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Case Definition

NOTE: Alberta Health will update this guideline as new information becomes available.

Preliminary Case

Children and adolescents 0–19 years of age with fever (38.1 °C or higher) for 3 days or longer.

AND two or more of the following:

- Rash or bilateral non-purulent conjunctivitis or muco-cutaneous inflammation signs (oral, hands or feet);
- Features of myocardial dysfunction, or pericarditis, or valvulitis, or coronary abnormalities (ECHO findings or elevated Troponin/NT-proBNP);
- Hypotension or shock;
- Evidence of coagulopathy (abnormal PT, PTT, elevated d-Dimers);
- Acute gastrointestinal problems (diarrhea, vomiting or abdominal pain);

AND

- Elevated markers of inflammation such as ESR, C-reactive protein or procalcitonin;

AND

- No other obvious microbial cause of inflammation, including bacterial sepsis, staphylococcal or streptococcal shock syndromes (e.g., respiratory panel, strep throat test, monospot, CMV IgM, EBV IgM, blood culture);

WITH OR WITHOUT^(A)

- Evidence of COVID-19 (RT-PCR, antigen test or serology positive) or likely contact with a person(s) with COVID-19.

^(A) This criteria is optional at this time in Alberta to ensure the case definition is sensitive enough to include all cases of concern. When the meaning and availability of serological testing is more completely understood this criteria may become a required criteria. Any information about COVID-19 testing results and any epidemiological link should be included in the appropriate sections of the revised *Alberta Multisystem Inflammatory Syndrome in Children and Adolescents (MIS-C) Report form*. The form is available at <https://www.alberta.ca/notifiable-disease-guidelines.aspx>.

Reporting Requirements

Physicians

- All physicians shall notify the Medical Officer of Health (MOH) (or designate) of the zone, of all MIS-C cases that meet the preliminary case definition by phone, fax or email within 48 hours (two business days) and include:
 - Name;
 - Age;
 - Date of birth;
 - Gender;
 - Personal health number;
 - Positive laboratory report (if applicable); and
 - Other relevant clinical/epidemiological information

NOTE: The revised *Alberta Multisystem Inflammatory Syndrome in Children and Adolescents (MIS-C) Report form* includes this information and can be used to notify the MOH. The form is available at <https://www.alberta.ca/notifiable-disease-guidelines.aspx>.

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- The MOH/designate of the zone where the case currently resides shall report all preliminary cases to the CMOH/designate using the *Notifiable Disease Report (NDR)* form or another mutually agreed upon reporting system (e.g., CDOM – *Rare or Emerging Communicable Disease*) within 7 days of initial notification as per usual.
- For out-of-province and out-of-country preliminary cases, the following information should be forwarded to the CMOH/or designate as per usual communicable disease reporting processes and include:
 - name,
 - date of birth,
 - out-of-province health care number,
 - out-of-province address and phone number,
 - any positive laboratory report/s, and
 - other relevant clinical / epidemiological information.

Epidemiology

Etiology

Multisystem Inflammatory Syndrome in Children and Adolescents (MIS-C) is a condition characterized by fever, inflammation, and multi-organ dysfunction that manifests late in the course of SARS-CoV-2 (COVID-19) infection.⁽¹⁾ Early data suggests that this syndrome is not the result of acute infection, but rather of a dysregulated immune response to the earlier infection or exposure.⁽²⁾

Clinical Presentation

Children and adolescents with MIS-C present clinically with some features similar to Kawasaki disease and toxic shock syndrome. Typically, it presents as an acute illness, accompanied by a hyper-inflammatory syndrome, leading to multi-organ failure, shock and rarely death. Children and adolescents with MIS-C may have persistent fever, fatigue, non-purulent rash, swelling of hands or feet, and a variety of signs and symptoms including multi-organ (e.g., cardiac, gastrointestinal, renal, hematologic, dermatologic, neurologic) involvement, and elevated inflammatory markers. Not all children and adolescents will have the same signs and symptoms.^(3–7)

Transmission

MIS-C is not communicable. However, if the child or adolescent tests positive for COVID-19 at time of presentation, they may still present a risk of transmission of COVID-19 to others.

Incubation Period

Presentation of MIS-C symptoms following COVID-19 infection range approximately 2-6 weeks following infection.^(2,8)

Period of Communicability

MIS-C is not communicable.

Host Susceptibility

Increased incidence observed in children and adolescents of African, Afro-Caribbean and Hispanic descent has been observed in the United States and United Kingdom.^(2,9)

Incidence

MIS-C became reportable in Alberta on May 27, 2020. In Canada, provinces and territories report cases of MIS-C to the Public Health Agency of Canada. However, data on incidence in Canada is not publicly available at this time. MIS-C is a rare (<1%) complication of COVID-19 infection in children and adolescents. Most cases tend to be in older children and adolescents who were previously healthy.⁽⁹⁾

Public Health Management

Diagnosis

The diagnosis of MIS-C is based on a combination of history, physical exam and laboratory tests that look for signs of inflammation if there is no other diagnosis.

- Tests aimed at identifying laboratory evidence of inflammation, cardiac or other multisystem involvement are warranted.
- Similarly, COVID-19 detection by RT-PCR or antigen test is indicated.
- Where feasible, COVID-19 serology testing is strongly recommended, especially if RT-PCR or antigen testing are negative. Any serological testing should be performed prior to administering IVIG or any other exogenous antibody treatments.

Key Investigation

- Confirm the diagnosis and that individual meets case definition.
- Obtain history of illness including date of onset of signs and symptoms.
- Determine any underlying chronic or immunocompromising conditions.
- Patient history (noting epidemiological link to known COVID-19 case or exposure.)

Management of a Case

- Consultation with pediatric specialists (e.g., rheumatology, immunology, infectious disease, cardiology) is recommended.
- Isolation precautions apply for hospitalized MIS-C cases with concurrent COVID-19 infection. Consult with hospital IPC for recommendations for lifting isolation.
 - Provide information about disease transmission and measures to minimize transmission, including practicing proper hand hygiene and respiratory etiquette.
- For information on infection prevention and control precautions refer to the following:
 - [AHS IPC Resources](#)
 - [Infection prevention and control for novel coronavirus \(COVID-19\): Interim guidance for acute healthcare settings](#)

Treatment

Treatments consist primarily of supportive care and are directed against the underlying inflammatory process.^(4,6,10,11)
Supportive measures include:

- fluid resuscitation;
- inotropic support;
- respiratory support; and
- in rare cases, extracorporeal membranous oxygenation (ECMO).

Anti-inflammatory measures have included the frequent use of intravenous immunoglobulin (IVIG) and steroids. The use of other anti-inflammatory medications and the use of anti-coagulation treatments have been variable. Antibiotics are routinely used to treat potential sepsis while awaiting bacterial cultures.

Management of Contacts

As recognition of this syndrome in children/adolescents possibly reflects past exposure to COVID-19, it is important to obtain a review the child/adolescent's history for any known or potential exposure to a person with COVID-19. Although this is not directly related to the management of the pediatric patient, it may have public health implications for other exposed persons who may benefit from testing.⁽¹²⁾

Preventative Measures

There is no known way to prevent MIS-C. However, the following COVID-19 preventive measures should be reinforced whenever in contact with the pediatric population or their parents:

- Avoid close contact with people that have acute respiratory infections,
- Maintain physical distancing (i.e., 2 metres/6 feet),
- Practice proper respiratory etiquette (i.e., cover coughs and sneezes with disposable tissues or clothing),
- Wash hands often with soap and water for at least 20 seconds,
- Avoid touching your face with unwashed hands,
- Avoid non-essential travel,
- Get immunized if appropriate for age group, and
- Watch for COVID-19 symptoms: cough, fever, shortness of breath/difficulty breathing, runny nose or sore throat.⁽¹³⁾

ANNEX A: Revision History

Revision Date	Document Section	Description of Revision
2020-05-27	Case Definition	Interim case definition adapted from WHO⁽¹⁴⁾ included.
2021-05-26	Case Definition	Added examples of viral and bacterial testing done to rule out other causes of illness
	Reporting	Updated requirements
	Management	New

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